LESSON PLAN

ADVANCED AIRCRAFT TECHNOLOGY (20)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	11
Total No. of Lesson Plans	20	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Concept of Glass cockpit and architecture, Ring Laser Gyros
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Concept of Glass cockpit and architecture, Ring Laser Gyros. Practical demonstration of Gyro properties like rigidity and precision.
5. Specific Objectives	20.4.1.1 Trainee will be able to explain principles of various advanced electrical systems.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Demonstration of Ring Laser Gyro properties (Rigidity and Precision) using model; Safety: Handle optical components gently; Supervision: Instructor monitors alignment and rotation.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of a gyroscope in aircraft navigation?
(b) How does a classical mechanical gyroscope differ from a Ring Laser Gyro?
(c) Define the term 'Glass Cockpit' in the context of modern aviation.

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Definition and scope of Glass Cockpit • Architecture of modern avionics integration • Benefits of digital over analog systems • Data bus architecture in glass cockpits • Redundancy and failure management systems	L & Ds	PPT, IB	15 min
(b)	• Operating principle of Ring Laser Gyro • Sagnac effect and light path interference • Construction of laser gyroscope components • Mechanism of rigidity in laser gyros • Sources of error and dithering techniques	L & Ds	PPT, PDF	20 min
(c)	• Demonstration of gyroscopic rigidity • Demonstration of gyroscopic precision • Comparison with classical mechanical gyros • Application in modern flight instruments • Maintenance and inspection requirements	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how the Sagnac effect enables the Ring Laser Gyro to measure rotation.
(b) Describe two key advantages of a Glass Cockpit over traditional analog instruments.
(c) Why is dithering necessary in the operation of a Ring Laser Gyro?

12. Home Assignment and Follow Up	Prepare a brief note comparing the architecture of a classical control system (covered in LP 12) with the data bus architecture of a Glass Cockpit, highlighting the shift from mechanical linkages to electrical signals.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Aero engine classical control system.	Refer to the master syllabus reference list	LP 12

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min